Hong Kong Baptist University

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GCAP3226 Group Project Report (Team 3)

Topic: Typhoon Signal Analysis

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I. Introduction (Hui Man Hei) /500-600

1.1 Research Context and Background

Recently, there were earlier typhoon signal announcements than traditional practices when typhoon was influencing Hong Kong. Take the latest super typhoon Ragasa as an example, Signal 8 was hoisted at 14:20 on 23 September 2025 (Hong Kong Time), although there were only 3 of 8 "selected due to their good exposure and geographical distribution of separation by Hong Kong mountain ranges" (Hong Kong Observatory (HKO), 2024) *(Image 1)* stations showed sustained speed exceed 63km/h in 10-minute interval (from HKO's API data). HKO then responded to that "too early" typhoon warning timing is key "forecast element" and function as "precautionary alert".

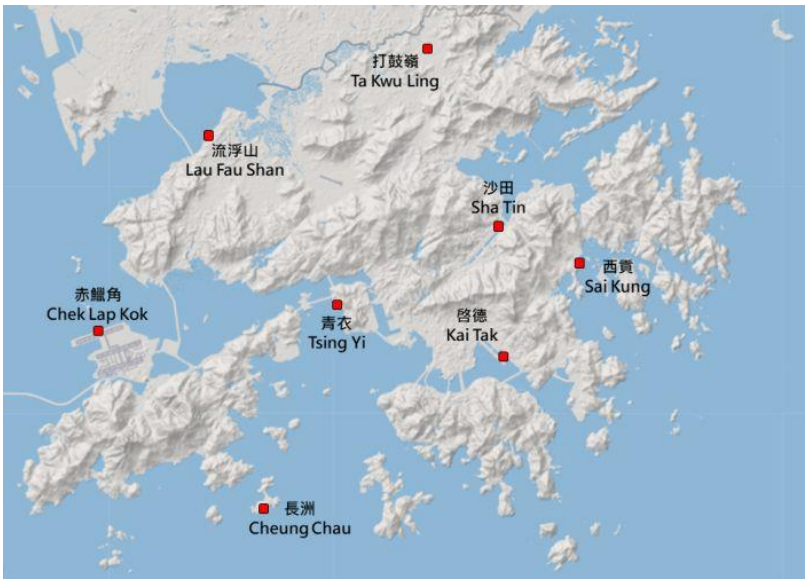


Image 1: 8 Weather Stations for Referenced Anemometers in the Tropical Cyclone Warning System (HKO, 2024)

Signal 8 decision timing has significant economic consequences for Hong Kong. According to Wang et al. (2024), Hong Kong Exchanges and Clearing Limited (HKEX) operated normally "despite the arrival of tropical storm Toraji leading to Signal 8 being issued on 13 November 2024 nighttime, which was the first time of operating during storm. This precedent "virtual trading and remote operations" (Wang et al., 2024) do not only show recent technological upgrades, but they also imply over-reaction by precautionary alerts. Hong Kong needs to suffer economic lost from passive shutdowns and disruptions from school suspension.

1.2 Research Objectives

1.2.1 Primary Objective

Improve transparency in the decision process by applying comprehensive real-time wind measurements analysis and official standards to access the accuracy of Signal 8 decisions.

1.2.2 Secondary Objectives

1. Analyze divergences between real-time wind data and Signal 8 decisions
2. Evaluate HKO standards transparency and decision-making procedures
3. Examine the correlation between announcement timing and decision accuracy
4. Develop evidence-based proposals for improving Signal 8 decision-making

1.3 Research Questions

1.3.1 Main Research Question

Do Signal 8 decisions in line with real time wind data and official standard and what factors contribute to inconsistencies?

1.3.2 Sub-questions

- Do Signal 8 notices align with official HKO criteria based on real time wind data?
- What discrepancies exist between real-time wind intensity and Signal 8 decisions?
- How transparent are HKO's criteria and decision-making processes?
- What factors contribute to accurate vs inaccurate Signal 8 decisions?
- How can data-driven analysis improve Signal 8 decision accuracy?

1.4 Significance and Expected Impact

It is important to use data-driven assessment with evidence-based evaluation using comprehensive data analysis. This is Hong Kong Baptist University students first study of Cross-validation of HKO's API real time data in 10 minutes interval (Website: Regional weather in Hong Kong – the latest 10-minute mean wind direction and wind speed and maximum gust | DATA.GOV.HK) and official notices. From studying existence of discrepancies of wind data, understanding of HKO's decision-

making processes is improved, it is owed to public access of wind data like data.gov.hk. In the long run, accessing accuracy of Signal 8 decision can facilitate policy improvements, providing advice for enhancing Signal 8 decision processes. For example, “2-Tier Signal 8” includes “Signal 8 Alert” and “Confirmed Signal 8”, the former needs to collect wind speed data to predict the potential or ultimate intensity and this alert should be effective before 4-6 hours; the latter needs to fulfill the wind intensity requirements (63-118 km/hour) in at least 4 weather stations, this can imply strong winds affect wide in Hong Kong and reduces unnecessary shutdowns. Moreover, accurate Signal 8 decision with consistence to wind speed data can build public trust, maintaining HKO credibility through transparent, data-driven decisions.

II. Literature Review and Theoretical Framework (Hong Kam Yin)

2.1 Signal 8 Decision-Making Literature

The issuance of a typhoon warning signal is a complex decision-making process. Within meteorological decision-making research, it is widely acknowledged that forecasters and officials operate under conditions of high uncertainty, balancing scientific data with judgment to manage public risk (Stewart, 2000). The Precautionary Principle, which advocates preventative action in the face of uncertain but potentially severe threats (Klinke & Renn, 2002), offers a key theoretical lens. This principle helps explain why a meteorological authority’s decision is not merely a passive reaction to data, but an active risk management practice aimed at preventing worst-case scenarios.

From a public policy analysis perspective, the Signal No. 8 decision is a focal point for government Transparency and Accountability. The literature consistently shows that transparency in how government bodies make decisions is directly linked to higher levels of public trust (Grimmelikhuijsen & Meijer, 2014). Furthermore, Stakeholder Theory is relevant, as it highlights the necessity for decision-makers to balance the often-competing interests of diverse stakeholders, including public safety, economic continuity, and transportation operations. Therefore, an evaluation of Signal 8 decisions must be situated within a broader socio-economic context, moving beyond a purely meteorological analysis.

2.2 Signal 8 Criteria and Standards

The core benchmark for assessing decision accuracy in this study is derived from the

official criteria established by the Hong Kong Observatory (HKO). According to official documentation, the No. 8 Gale is issued when sustained winds of 63 to 117 km/h. The decision is primarily based on a network of eight reference anemometers, with the criterion met when gales are recorded or anticipated at "half or more" of these stations (Hong Kong Observatory, 2007). This clearly defined, quantitative standard provides the objective "yardstick" for this research's data-driven assessment.

In comparison, other international systems, such as the Saffir-Simpson Hurricane Wind Scale used by the U.S., focus primarily on a storm's absolute intensity based on its maximum sustained winds (National Hurricane Center, 2023). Hong Kong's system, in contrast, is more focused on the storm's widespread impact on the local territory. This distinction underscores the localized and impact-based nature of effective warning systems. Consequently, this study will strictly adhere to Hong Kong's official local criteria as the standard for evaluation.

2.4 Research Gaps and Contributions

Although some news media and members of the public have discussed meteorological decision-making theories and the official criteria of the HKO, there is a distinct lack of systematic, empirical research that uses high-frequency, real-time wind data to quantitatively assess the accuracy of Signal No. 8 decisions. Previous discussions have often focused on qualitative analyses of individual typhoon cases, failing to provide objective, data-driven evidence across multiple typhoons.

Therefore, this project’s primary contribution is to fill this gap. By developing and applying a comprehensive, data-driven framework, this study will conduct a novel, evidence-based evaluation of Signal 8 decision accuracy. It will offer an empirical analysis of the alignment between real-time weather conditions and warning issuances, providing valuable insights for enhancing the transparency, accountability, and effectiveness of Hong Kong’s typhoon warning system.

III. Methodology and Data Collection

IV. Data Analysis and Results (Zheng Zian)

V. Findings and Implications (Chen Man Ching)

5.1 Key Findings

5.1.1 Data Discrepancy Findings

Our analysis of 11 typhoons between 2022 and 2025 shows clear discrepancies between real-time wind data and the Hong Kong Observatory's (HKO) Signal No. 8 decisions. Among the cases studied, only Typhoons Saola and Ragasa fully met the official requirement where at least four out of eight reference stations record sustained winds at or above the threshold. For most other typhoons, the signal was issued before these conditions were met or when fewer than four stations reached the required speeds. Notably, the lead times for Saola and Ragasa were extremely long, issued approximately 1,000 minutes (16.7 hours) and 700 minutes (11.7 hours) in advance respectively. This far exceeds the ideal 3–4 hour preparation window, indicating that actual timing often prioritizes caution over strict adherence to station data.

5.1.2 Decision-Making Process Analysis

The decision-making process appears to incorporate more than just wind speed. HKO considers forecast uncertainty, coastal exposure, expected gust patterns, and the potential consequences of issuing the signal too late. This precautionary approach enhances public safety but may cause the signal to deviate from strict numerical criteria, leaving the public unclear about why certain warnings are issued.

5.1.3 Government Response Assessment

Although real-time wind data is publicly accessible, the reasoning behind Signal No. 8 decisions is not clearly explained. The public can see the wind observations, but they cannot understand why certain decisions were made, especially in situations where the wind speeds at the reference stations did not reach the required threshold.

The lack of clear explanation for these borderline decisions creates a communication gap and reduces transparency. This makes it difficult for the public to evaluate whether the decisions were justified, even when safety was the primary concern.

5.2 Implications for Policy and Practice

5.2.1 HKO Decision-Making

These findings suggest that the Hong Kong Observatory needs to improve how it balances public safety and social disruption. Because a Signal No. 8 warning can bring the city to a stop, every decision must be well justified. While issuing the signal very early can enhance safety, this approach may not be ideal for a major financial and economic centre. The decision-making process should therefore aim for a better alignment between forecast confidence, public risk, and the level of disruption caused.

5.2.2 Public Policy Impact

The economic consequences of these decisions are substantial. Our analysis indicates that direct economic losses ranged from approximately HKD 140 million during Typhoon Koinu to HKD 1.74 billion during Typhoon Haikui and its aftermath. Earlier signal issuance prolongs citywide shutdowns, thereby increasing costs. These losses stem from both physical storm damage and the suspension of commercial activities. Beyond economics, the signal disrupts daily life by halting transport, schools, and hourly work. This highlights a critical socio-economic trade-off that requires careful calibration to protect safety while minimizing unnecessary paralysis of the city.

5.2.3 Academic and Research Contributions

This study acts as an independent review by analyzing data from 11 typhoons. By comparing official records against the rules, we provide the public with an unbiased look at whether the No. 8 signal was truly justified. This research improves transparency by showing how open data can be used to see if the government is actually following its own criteria.

5.3 Recommendations for Improvement

5.3.1 Enhancing the Decision-Making Process

The HKO should make its decision process more transparent by clearly explaining the official criteria and how supplementary factors influence decisions when wind data falls short. Implementing a public decision log that records key data, reasons for early issuance, and forecast confidence levels would bridge the communication gap. Whenever feasible, the Observatory should aim for a standard 3–4 hour lead time

before gale-force winds arrive to avoid excessive disruption.

5.3.2 Strengthening Government Accountability

To enhance accountability, the government should conduct systematic post-typhoon reviews and make these findings public. The decision-making process would benefit from regular consultation with emergency services, transport operators, and community stakeholders. Furthermore, an accessible public enquiry mechanism is necessary to ensure citizens can raise concerns and receive well-substantiated responses regarding warning decisions.

5.3.3 Supporting Research and Ongoing Evaluation

Finally, continuous evaluation is essential for system improvement. This involves regularly verifying whether Signal No. 8 timing aligns with actual wind conditions and updating criteria based on empirical evidence. The government should invest in advanced forecasting tools to better estimate uncertainty. By periodically reviewing the system and incorporating data-driven updates, the warning mechanism can remain reliable and responsive to public needs.

VI. Conclusion and Future Research (Kung Tsz Lok)

6.1 Key Findings Summary.

Overall, our research provided a comprehensive and data-driven assessment of the accuracy Signal 8 decisions through the integration of the real time wind data. Our analysis reveals a discrepancy between actual wind conditions and the issuance of Signal 8 warnings which suggest an inconsistency in the decision-making framework. Another critical issue found was the limited transparency surrounding how these decisions were made while public access is restricted to get relevant information. These findings suggest the need for clearer communication and stronger accountability within the warning system. In view of these gaps, our study proposed serval evidence-based policy recommendations aimed at improving the objectivity and public confidence in weather warning process.

6.2 Research Contributions

The study contributes to a field which introduces an innovative methodology for assessing the accuracy of weather warnings. By integrating of all real time wind data, it made a systematic approach to evaluating the performance of decision-making

system used during different weather events. Our research also sheds light on the intersection from meteorological science and governance which emphasizes the importance of transparency and accountability in public emergence management. Evidence-based recommendations generated from this study provide practical insights which could inform policy reforms and support the development of more responsive and decision made based on data framework.

6.3 Limitations and Challenges

While the contribution has made a huge impact, our study reveals different limitations. First is the accessibility to historical Signal 8 decision data which limit the scope of analysis to more recent incidents. Moreover, the decision-making process itself is a complex which needs to involve multiple meteorological and admin factors that are not always disclosure to public. Consequently, despite our findings offering valuable insights, they represent only a part of ice, it needs more fact that require further investigation.

6.4 Future Research Direction

It is suggested that future research should adopt a longitudinal approach which extend the timeframe of analysis to find long term trends and patterns in decision of making Signal 8. Also, there is also a need to develop and refine decision making frameworks, that could consider both quantitative data and specific field's expert judgement to make it more effectively. Further exploration in mechanisms which can enhance transparency and accountability such as open data platforms like data.gov.hk can also build citizens trust in the system. Finally, we should collaborate among expert like meteorologists, data scientists and policymakers can promote more adaptive, policies based on evidence.

6.5 Contribution to Sustainable Development Goals (SDGs)

Our research contributes directly to different Sustainable Development Goals (SDG). For SDG 11, which is sustainable cities and communities, our study improved disaster prepares and urban resilience through more accurate and timely weather warnings. Meanwhile, SDG 13 which is Climate Action can deal by promoting adaptive strategies to strengthen climate resilience. For SDG16, (peace, justice and strong institutions_, if we can make a better transparency and accountability in decision making, SDG 16 would be benefited. Finally, for SDG 17 (Partnership for the goals), the emphasis on collaboration across sectors and discipline can reinforcing the importance of co-operative governance in achieving sustainable and resilient societies.

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